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Thm 7:

Rings and semirings

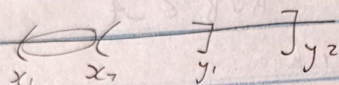
$$1. \quad A, B \in \mathcal{G} \Rightarrow A \setminus B \in \mathcal{G}^+, \quad \mathcal{G} = \{ (x, y) : x, y \in \mathcal{C} \}$$

$$\mathcal{G}^+ = \{ \sum A_i : A_i \in \mathcal{A}, A_1, \dots, A_n \text{ p.d. } n \in \mathbb{N} \cup \{0\} \}$$

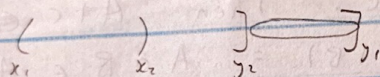
Proof: Let $A = (x_1, y_1]$, $B = (x_2, y_2]$.

Case 1: $x_1 < x_2 < y_1 < y_2$

$$A \setminus B = (x_1, x_2] \in \mathcal{L} \subseteq \mathcal{G}^+$$

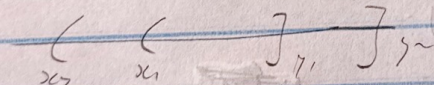


(Similarly, if $x_1 < x_2 = y_2 < y_1$,



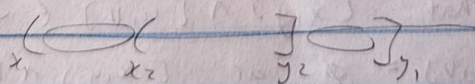
$$A \setminus B = \{ (y_2, y_1] \} \in \mathcal{L} \subseteq \mathcal{G}^+$$

Case 2: $x_2 < x_1 < y_1 < y_2$



$$A \setminus B = \emptyset \in \mathcal{L} \subseteq \mathcal{G}^+$$

Case 3: $x_1 < x_2 < y_2 < y_1$

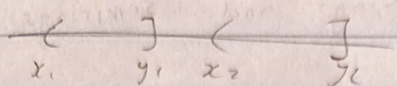


$$A \setminus B = \{ [x_1, x_2] \cup (y_2, y_1] \}$$

$\Rightarrow (x_1, x_2]$, $(y_2, y_1]$ disjoint, $(x_1, x_2]$, $(y_2, y_1]$ disjoint

$$\Rightarrow (x_1, x_2] \cup (y_2, y_1] \in \mathcal{G}^+$$

Case 4: $x_1 < y_1 < x_2 < y_2$ (or $x_2 < y_2 < x_1 < y_1$)



$$A \cap B = [x_1, y_1] \cap [x_2, y_2] = [y_1, x_2] \in \mathcal{I} \subseteq \mathcal{I}^+$$

to prove: every ring is a semiring

Proof: Let $\mathcal{A} \subseteq \mathcal{P}(X)$ be a ring

1. $\emptyset \in \mathcal{A}$
2. $A, B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A}$
3. $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$

To show it is a semiring

1. $\emptyset \in \mathcal{A}$ (trivial)
2. $A, B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A}$

Let $C = (A \setminus B) \in \mathcal{A}$. Then $A \cap C \in \mathcal{A}$,
and $A \setminus C = A \setminus (A \setminus B) = A \cap B \in \mathcal{A}$. Then
 $A \cap B \in \mathcal{A}$.

3. $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}^+$ By 2.

for a ring, $A, B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A}^+$
 $\therefore \mathcal{A}$ is a semiring.

$\mathcal{A} = \{[a, b] : a, b \in \mathbb{R}\}$. Not closed under union,
 $[1, 2] \cup [3, 4] \notin \mathcal{A}$, but it is a semiring.

1. $\emptyset \in \mathcal{A}$ (take $a=b$) (x, x)

2. $[x, y], [u, v] \in \mathcal{A}$

We need to show that $[x, y] \cap [u, v] \in \mathcal{A}$. We
know that $[x, y] \cap [u, v] = \emptyset$ or $[x, y] \cap [u, v] =$
 $[\max\{x, u\}, \min\{y, v\}]$

3. We need to show that for $[x, y], [u, v] \in \mathcal{A}$,
 $[x, y] \setminus [u, v] \in \mathcal{A}^+$. However, the difference of 2
intervals is a finite union of disjoint intervals.

3.a $\{A_\beta\}_{\beta \in B}$ be a family of rings. We need to prove $\bigcap_{\beta \in B} A_\beta$ is a ring. Let $\bigcap_{\beta \in B} A_\beta = M$.

1. $0 \in M$

Since A_β is a ring, $\forall \beta \in B$, $0 \in A_\beta \Rightarrow 0 \in M$

$\therefore A, B \in M \Rightarrow A+B \in M$.

Let $A, B \in M$. For each $\beta \in B$, $A, B \in A_\beta$ and since A_β is a ring, $A+B \in A_\beta$ and $A \cdot B \in A_\beta$. So, $A+B \in M$ and $A \cdot B \in M$. So M is a ring.

b. Suppose $X = \{1, 2, 3, 4\}$. Let $S_1 = \{\emptyset, \{1\}, \{1, 2, 3, 4\}, \{1, 3\}, \{4\}\}$, $S_2 = \{\emptyset, \{1\}, \{1, 2, 3, 4\}, \{3, 4\}, \{2\}\}$. Both are semirings but their intersection $\{\emptyset, \{1\}, \{1, 2, 3, 4\}\}$ which is not a semiring.